

CONFLUENCE ENGINE PROJECT · BUILDER
PACKAGE

VCE-EQC-001
Revision
1.0
2026

Engineer's *Quick Check*

VOLTA CONFLUENCE ENGINE — A ONE-SITTING
FEASIBILITY READ BEFORE YOU OPEN THE FULL GUIDE

PART 1

Does the physics hold up?

PART 2

Is it worth building?

LICENCE

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This is a one-sitting feasibility read for an engineer. The first half asks the only question that matters up front — *does the physics hold up well enough to be worth anyone's time?* The second half is the cost, time, and skill reality: *if it works, is it worth building?* Every claim carries a certainty label so you can see at a glance where the ground is solid and where it is genuinely untested.

Read this first. This design was conceived by an artist, not an engineer, and documented with

discipline — but not an engineer's instincts.

Considerable due diligence has gone into it, and every figure is labelled for what it is: CONFIRMED physics or catalogue fact, CALCULATED from those, or ESTIMATED engineering judgement. What the author cannot do is check the work the way you can. That is the entire purpose of this document: to put the reasoning in front of someone qualified to find the flaw. If the physics does not hold, this project wants to know before anyone spends money — please say so.

PART 1 · THE PHYSICS

Can It Get Off *the Ground*?

1 · THE ENERGY BALANCE — THE MAKE-OR-BREAK NUMBER

Everything hinges on one question: can a human hand-crank actually supply the power the instrument draws, continuously, through a performance? This is the tightest margin in the design and the first thing to check.

Power available

~31 W sustained one-hand crank (95% of population; literature reports 54 ± 14 W for young adult males)

CONFIRMED

Power required

18-23 W continuous system draw CALCULATED

Margin

~31 W against ~20 W. Positive, but modest — roughly 1.3-1.7×, not a comfortable multiple.

The energy is buffered so cranking and playback need not be simultaneous: the generator produces 24 V AC, a

passive Cockcroft-Walton multiplier steps it to 600 V DC, and a ceramic capacitor bank stores 4,000–5,000 J.

CALCULATED At 20 W draw a full bank yields ~250 seconds (~4 minutes) of play, so the operator re-cranks every 3–4 minutes. **CALCULATED** Ceramic dielectric self-discharge is negligible over performance timescales, which is why a capacitor bank is used instead of batteries or supercapacitors. **ESTIMATED**

THE ONE THING MOST LIKELY TO SINK IT

The 31 W figure is comfortable cranking in isolation. Nobody has measured whether that margin survives an operator who is *also* concentrating on a technically demanding DJ set for 45+ minutes — cognitive load and fatigue both eat into available power. This is the design's **single largest open question**, and it cannot be resolved on paper. If you doubt one thing, doubt this — and it is exactly what first-prototype Experiment 1 is built to measure.

2 · THE SIGNAL CHAIN — THE PART YOU CAN VERIFY IN MINUTES

This is the most conventional element of the whole instrument: a standard two-stage valve phono preamplifier with passive RIAA, feeding transformer volume controls. The gain arithmetic is checkable against ordinary practice.

Stage Element		Gain / effect
1	MC cartridge	0.2–0.5 mV output
2	Step-up transformer (Lundahl LL1931, 1:10)	+20 dB
3	12AX7 triode, first stage	+26 dB

4	Passive RIAA network (IEC 60098)	-20 dB @ 1 kHz
5	12AX7 triode, second stage	+26 dB
Net cartridge → line		~+52 dB → 300-500 mV line level
CALCULATED		
6-8	Transformer volume controls + equal-power crossfader (passive)	Attenuation only; ~0 added noise
9-10	Output transformer → field coil compression drivers + horn	Impedance match → acoustic output

The arithmetic: +20 – (RIAA insertion) + 26 – 20 + 26 ≈ +52 dB, lifting a ~0.3 mV cartridge signal to line level. Placing RIAA *between* the two gain stages (not at the input) is deliberate — it keeps the first stage's noise performance intact. All of this is textbook valve-phono engineering; if something here is wrong, it should be quick for you to spot. CONFIRMED — STANDARD PRACTICE

3 · WHY 20 WATTS CAN FILL A ROOM

The apparent paradox — concert volume from a hand-crank — resolves at the speakers. The field coil compression drivers on a corner-loaded horn have a sensitivity of **108–112 dB/W/m** CONFIRMED — DRIVER CLASS, among the highest achievable. At that efficiency, single-digit acoustic watts produce 100–112 dB SPL average in a suitable room. ESTIMATED The whole instrument is an exercise in efficiency: high-sensitivity horns are what make the tiny power budget viable, and they are the reason the energy balance in Section 1 is merely tight rather than impossible.

4 · THE REMAINING PHYSICS, AT A GLANCE

Element	Principle	Status
Remontoire spring drive	Constant-force turntable drive, <2% torque variation (Harrison H2 precedent, 1739)	CALCULATED
Magnetic bearing	Neodymium levitation; target stray field <5 µT at cartridge	ESTIMATED
Eddy-current governor	Passive speed regulation (Foucault braking)	CALCULATED
Cockcroft-Walton multiplier	Passive 24 V AC → 600 V DC ladder (1932)	CALCULATED
Passive HV distribution	Resistive dividers + zener refs; no switching noise; 15-25% resistive loss	ESTIMATED

5 · WHERE THE PHYSICS IS GENUINELY UNTESTED

Five bench experiments are specified to convert estimate into measurement. Running them first would raise first-prototype confidence from 88% to an estimated 93-95%. They are, in priority order:

# Experiment	Resolves
1 Generator output under a simulated 90-minute performance with deliberate cognitive load	The energy-balance question above — the big one
2 Spring motor run-time, full wind, representative platter load, 5 consecutive cycles	Whether one wind truly covers an LP side

3	Mu-metal shielding, gaussmeter at cartridge with bearing energised	Whether magnetic hum stays below the noise floor
4	Crossfader equal-power law and channel isolation, 1 kHz	Whether the passive mix behaves
5	Full signal-chain noise floor, A-weighted	Whether the system is quiet enough

PART 2 · THE COST / BENEFIT

Is It Worth *Building*?

6 · WHAT YOU NEED TO BRING

Capability	Why / note
High-voltage electronics	Mandatory. The bank stores 600 V DC. This is qualified-technician territory, not hobby soldering.
CNC / machine-shop access	Platters, brass cups, governors, tonearm parts.
Clockmaker commission	The two remontoire spring motors. Longest-lead item and hardest to source.
Field-coil speaker builder	Compression drivers + woofer, custom commission.
Cabinetmaking	Corner horn, plinths, console — walnut and brass.

7 · TIME AND MONEY

Build time

300–500 hours over 6–12 months
(experienced team) ESTIMATED

Total cost
(Option A) **\$33,912 – \$79,608 USD** ESTIMATED

Longest lead Remontoire spring motors — 8–20
weeks. Order first, before anything else.

Biggest Spring motors \$6–14k (pair) · field coil
single costs drivers \$3–8k · horn enclosure \$3–6k

8 · CONFIDENCE

First-prototype confidence **88%**
ESTIMATED — ENGINEERING
JUDGEMENT

After running the 5
experiments **~93–95%** ESTIMATED

Second-prototype
confidence **93%** ESTIMATED

9 · THE HARD LIMITS YOU CANNOT ENGINEER AWAY

These are not defects to be fixed — they are the instrument's fundamental character. Accept them or walk away.

F1 The operator supplies *all* energy, continuously —
~20 minutes of cranking per 2-hour performance.

F2 Valve warm-up: 60–90 seconds before play can
begin.

F3 Spring motor rewind between LP sides — ~25–28
minutes of run per wind.

F4 The corner horn needs a real 90° room corner to
load correctly — a venue constraint.

Five engineering problems are identified and each has a documented mitigation — worth reviewing in full in the complete guide (Section XI): 600 V safety (E1, contained by enclosure), generator-to-cartridge EMI (E2, mu-metal shielding), the quartz PLL being post-Victorian technology (E3, a philosophical rather than functional issue), unipivot tonearm azimuth variation (E4), and field-coil supply sensitivity (E5).

Build it if...

- You have or can commission high-voltage, machining, clockmaking, speaker, and cabinet skills
- You can absorb \$34k–\$80k and 6–12 months
- You accept F1–F4 as the instrument's nature, not flaws
- You are drawn to being the first to prove (or disprove) the concept in the real world

Walk away if...

- You need a working DJ tool rather than an engineering project
- High-voltage work is outside your competence and you can't partner for it
- The tight energy margin (Section 1) is a dealbreaker until measured
- You cannot commit the lead time for the spring-motor commission

If you found a flaw: that is a successful use of this document. The design is open (CC BY-NC) precisely so that qualified eyes can catch what a non-engineer could not. Corrections, doubts, and "this number is wrong because..." are all welcome through the project links below and feed directly into the next revision.

INSTRUMENT

Volta
Confluence
Engine
Human-
Powered
Two-Channel
Phonograph
Mixer

DOCUMENT

Engineer's
Quick
Check
VCE-
EQC-001 ·
Revision
1.0
2026

FULL GUIDE

See volta-
confluence-
engine-
complete for
all
specifications,
drawings, and
derivations

ONLINE

Open Source:
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